

Quantifying Interpretive Deviations in AI-Generated 3D Architectural Models

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ABSTRACT

This paper investigates image-to-3D reconstruction as a form of algorithmic spatial translation rather than as a neutral process of geometric restitution. Moving from the premise that every act of representation constitutes a mediated translation, the study examines how generative AI systems reorganize architectural spatial relationships when inferring three-dimensional form from a single photorealistic image. A complex architectural case study, the Perugia Fontivegge railway station, provides a controlled framework in which a complete and metrically verified digital model is available as reference. The availability of the original model enables a direct comparison between explicit design configuration and AI-generated reconstruction. An open-source workflow was implemented, testing multiple image-to-3D systems and selecting the most volumetrically coherent output for quantitative and morphological evaluation. Metric comparison was structured at two levels: a global assessment of the entire reconstructed mesh and a localized analysis restricted to the façade visible in the input image. The results reveal a measurable contraction of geometric deviations in the observable portion compared to the globally inferred model, confirming that larger discrepancies are concentrated in areas reconstructed without direct visual information. Where visual data are available, spatial relations remain relatively stable; where absent, volumetric configuration is reorganized according to probabilistic logics. Rather than interpreting these deviations solely as errors, the paper reads them as quantitative traces of inferential processes. Image-to-3D reconstruction thus emerges as a negotiated configuration between design intent and algorithmic synthesis. The study proposes an interpretative framework for assessing the epistemic limits and operative potential of AI-generated 3D models in architectural and heritage-oriented representation.



Introduction

Tra Every act of representation translates rather than reproducing. An image does not coincide with the object it describes; rather, it constitutes a translation regulated by instruments, conventions, and technical procedures (Goodman, 1976; Hodder, 1999). With the digitization of design, this translation has progressively become computational: geometry is no longer merely drawn, but constructed through formally explicated and controllable operations (Carpo, 2017; Dussere & Mitchell, 1983;). The introduction of generative artificial intelligence models further transforms this framework. Unlike deterministic systems, neural models learn structures and correlations from large datasets and operate through processes of statistical inference (Goodfellow et al., 2016). Recent architectures based on diffusion models have significantly expanded generative capabilities, enabling the production of images with high visual coherence from textual inputs (Chang et

al., 2023; Dehouche & Dehouche, 2023). Building on these developments, generative AI has progressively articulated itself in visual transformation workflows such as image-to-image (Rombach et al., 2022; Zhang et al., 2023; Xia et al., 2025), image-to-video (Ho et al., 2022; Blattmann et al., 2023; Zheng et al., 2024), and three-dimensional reconstruction processes (Poole et al., 2022; Liu et al., 2023; Tochilkin et al., 2024), in which the image becomes an operative material capable of generating continuity, variation, and reinterpretation.

In particular, image-to-3D procedures extend the generative paradigm to the spatial domain, transforming a two-dimensional view into a volumetric configuration. The possibility of reconstructing a spatial model implies that depth, proportions, and relationships are inferred from partial two-dimensional information. Research on monocular depth estimation (Bianconi et al., 2024; Eigen et al., 2014; Godard et al., 2017) and three-dimensional reconstruction (Choy et al., 2016; Mild-

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enhall et al., 2020; Qin et al., 2024; Wu et al., 2016) has demonstrated the technical feasibility of this transition, while also highlighting its fundamentally inferential character. More recent advances in generative 3D modeling, particularly approaches integrating diffusion models and neural radiance fields, have further consolidated this trajectory (Kerbl et al., 2023; Meshy AI, 2023). However, the majority of these studies evaluate model performance in terms of accuracy against standardized datasets or perceptual coherence (Tatarchenko et al., 2015; Tulsiani et al., 2017), while the issue of the stability of spatial relationships when reconstruction is applied to complex and design-defined architectural configurations remains comparatively underexplored.

In the field of architectural representation, this condition assumes particular relevance. The quality of a model cannot be assessed solely in terms of perceptual plausibility; it requires attention to the stability of spatial relationships and the volumetric consistency of the architectural organism (Carpo, 2011; Evans, 1995; Tulsiani et al., 2017; Girardeau-Montaut, 2016). Unlike many computer vision applications, where visual plausibility may suffice, in design representation metric correctness and relational coherence acquire central epistemic and operational significance (Chaillou, 2019; del Campo & Manninger, 2022; Bono, 2024).

When three-dimensionality does not derive from an explicit modeling process or from metric survey, but rather from learned estimation, it becomes necessary to interrogate the relationship between the design configuration and the generated configuration. Three-dimensional reconstruction thus opens a critical field of inquiry in which the difference between the original and the reconstructed configuration does not merely represent a geometric deviation; rather, it becomes a trace of the ways in which algorithmic inference organizes and synthesizes architectural complexity.

Aims and objectives

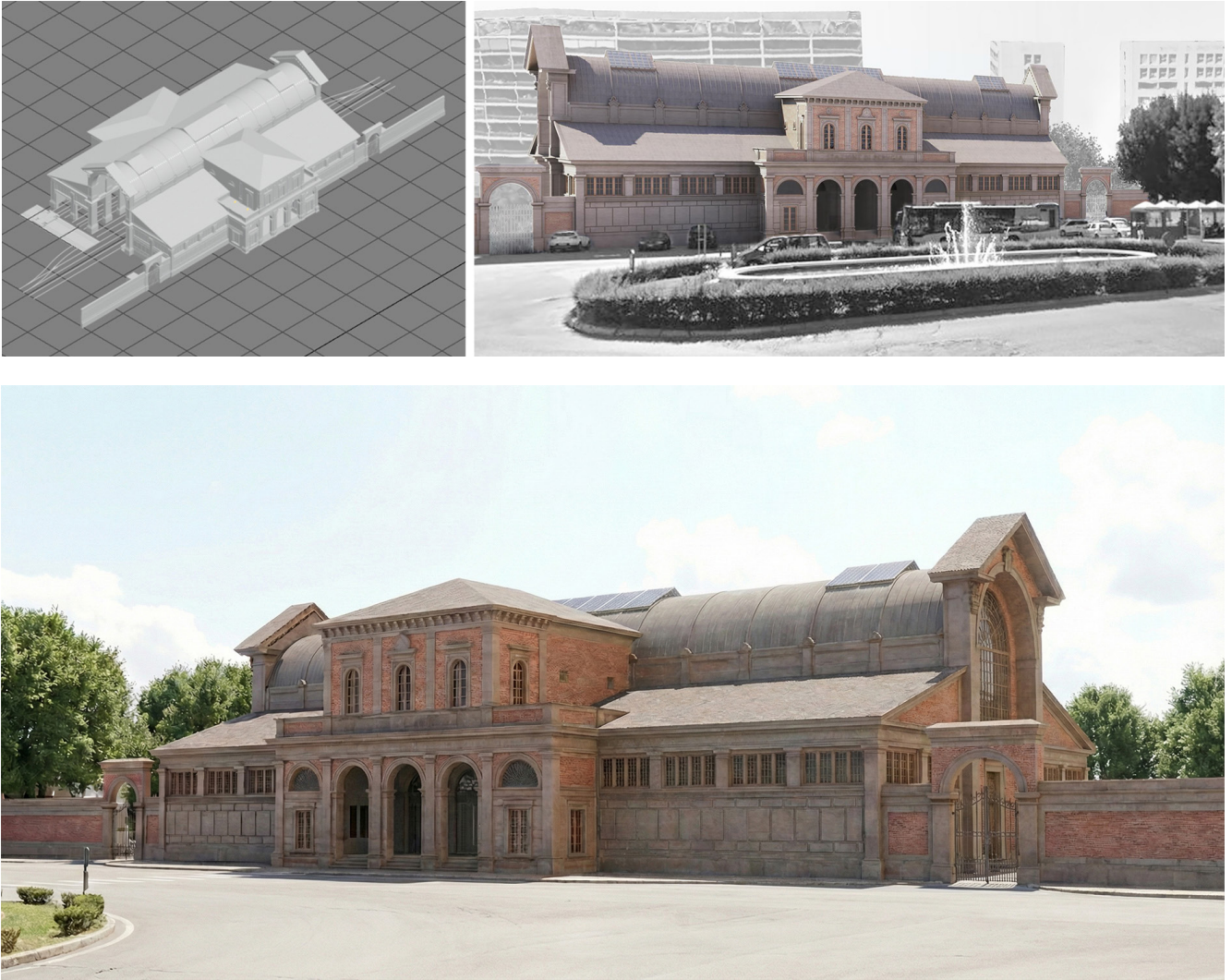
The present contribution therefore aims to critically investigate the inferential behaviour of an image-to-3D workflow, examining its capacity to generate a coherent three-

dimensional reconstruction from a single photorealistic image of an architectural project for which the original digital model is available. The objective is neither to measure system performance in general terms nor to propose a comparative evaluation against traditional modeling or surveying methods. Rather, it seeks to understand how spatial structure is reorganized in the transition from an explicit design configuration to its algorithmic translation.

The availability of the original three-dimensional model, complete and geometrically defined, makes it possible to treat the automatic reconstruction as the object of a controlled comparison, not only at the visual level but also in terms of volumetric configuration and proportional relationships among parts. The comparison between the original and the generated model allows for a systematic observation of the transformations introduced by the inferential process, isolating differences not as mere inaccuracies but as structural indicators of the system's operative logic.

The research consequently aims to quantify the magnitude and distribution of geometric deviations through comparative metrics, interpreting these differences as operative traces of the ways in which the algorithm deduces depth, proportions, and spatial relationships from partial two-dimensional information. Established approaches to geometric comparison in computer vision and 3D reconstruction, such as Chamfer Distance and Earth Mover's Distance, as well as intersection-over-union and volumetric consistency measures, provide methodological references for structuring this evaluation, while remaining secondary to the interpretative intent of the study.

In this sense, the study seeks to delineate an interpretative framework capable of making explicit the conditions of applicability, margins of reliability, and epistemic limits of image-to-3D techniques within the domain of architectural representation. Rather than offering a purely performance-based assessment, the contribution proposes a critical reading of three-dimensional reconstruction from images as a form of algorithmic spatial translation,



highlighting the implicit logics that govern the transformation from image to volumetric configuration.

Methods and procedure

The selected case study is the Perugia Fontivegge railway station (Bianconi et al., 2022). The project is supported by a complete three-dimensional model, developed and verified within a digital modeling environment and adopted here as the metric and morphological reference for the investigation.

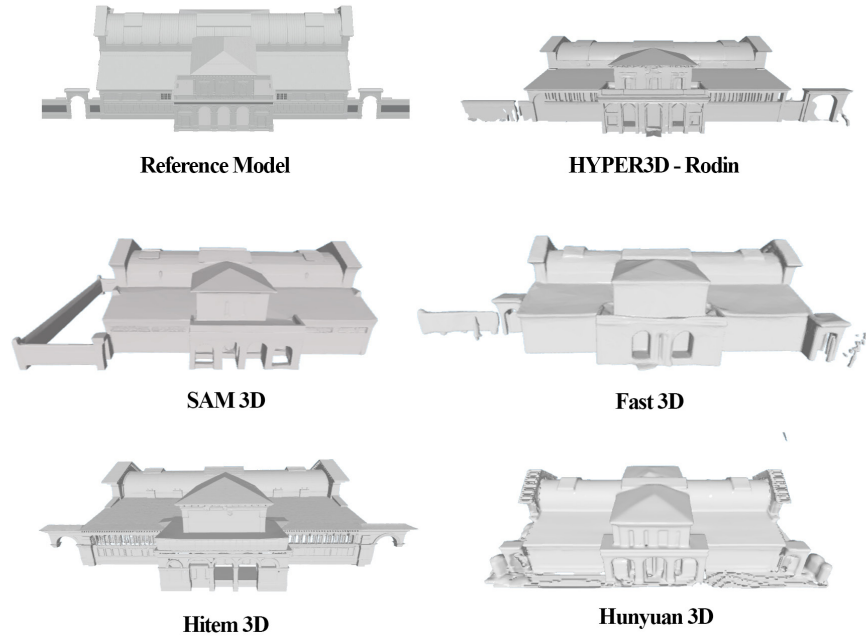
The choice is deliberate rather than incidental. The architectural organism exhibits significant volumetric complexity, characterized by articulated spatial configurations, overhangs, systems of openings, and hierarchical relationships among its constituent parts (Fig. 1). Such a configuration enables the testing of image-to-3D systems

under non-simplified conditions, where geometric restitution cannot be reduced to elementary volumes but must instead engage with a spatially articulated and compositionally structured organization. Starting from the original model, a single photorealistic rendering was produced and used as the sole input for the entire reconstruction process (Fig. 2). The choice of a rendered image, rather than a real photograph, responds to the need to control the initial dataset. The synthetic image eliminates contingent and non-governable variables, such as optical distortions or environmental interferences, ensuring full coherence between the representation and the reference model. In this way, any deviations observed in the reconstruction can be attributed to the inferential behavior of the system rather than to imperfections in the source material. No additional views or supplementary three-

Fig. 1 - Left: original three-dimensional digital model of the Perugia Fontivegge railway station, adopted as the metric and morphological reference for the study. Right: photorealistic image of the architectural project. Source: elaboration of Chiara Mommi.

Fig. 2 - Photorealistic rendering generated from the original three-dimensional digital model of the Perugia Fontivegge railway station and used as the sole input for the image-to-3D reconstruction workflow. Source: elaboration of Chiara Mommi.

Fig. 3 - Comparative visualization of the three-dimensional models generated from a single input image. Source: elaboration of Chiara Mommi.



dimensional information were provided. The experiment was conducted following an open-source approach, prioritizing platforms that allowed unrestricted export of the generated meshes. Several single-image 3D reconstruction systems were preliminarily tested, including *Meta AI's SAM 3D* (Wang & Zhang, 2024), *Hyper3D - Rodin Gen 2 Edit v1.9* (Wang et al., 2023), *Hitem3D v2.0* (Weiss & Beskardes, 2025), *Fast 3D* (Tochilkin et al., 2024) and *Hunyuan3D v2.0* (Hunyuan3D et al., 2025), all implemented through the *ComfyUI* platform (ComfyUI, n.d.). Each system was used under standard configurations, avoiding any subsequent manual intervention that might alter the originally inferential nature of the produced geometry.

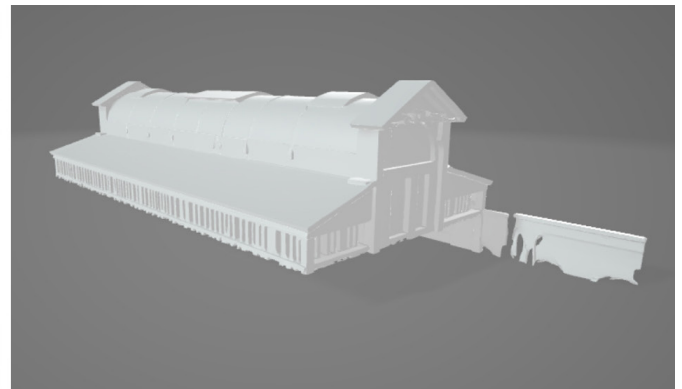
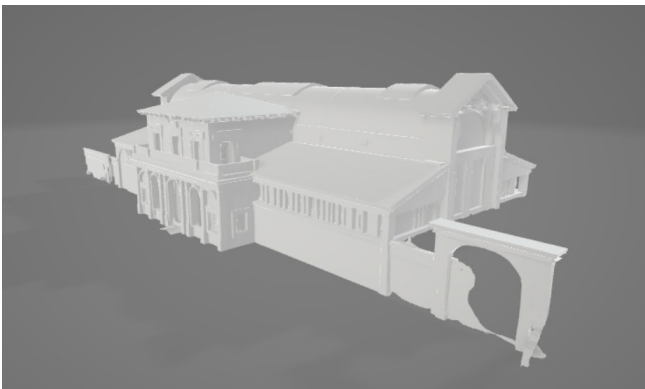
The preliminary evaluation of the results (Fig. 3) indicated that the model generated with *Hyper3D - Rodin* exhibited greater overall volumetric coherence, particularly in the definition of openings and in the stability of the principal proportional relationships. The other systems demonstrated more pronounced tendencies toward simplification or structural distortions, reducing the relevance of an in-depth metric analysis. For this reason, the quantitative comparison was conducted exclusively on the model produced with *Hyper3D - Rodin*, adopted as a representative case for verifying the reconstructive capacity

of image-to-3D systems when applied to a complex architectural configuration.

The generated meshes were exported in a standard format and imported into *CloudCompare* (Girardeau-Montaut, 2016) for metric analysis. The reconstructed model was first scaled relative to the original reference model and subsequently aligned through rigid registration using the Iterative Closest Point (ICP) algorithm, maintaining the design model as the invariant reference. A Cloud-to-Mesh distance computation was then performed in order to analyze the distribution of deviations between the corresponding surfaces.

To avoid an oversimplified interpretation of the metric data, the comparison was structured on two distinct levels. In the first phase, a global comparison was conducted across the entire mesh, including both the portions directly visible in the input render and the rear or occluded areas that were necessarily inferred by the system and therefore structurally different from the reference model (Figs. 4, 5). This measurement provides an overall mean value of geometric deviation and allows an assessment of the general behavior of the generative model.

In a second phase, the analysis was restricted to the frontal portion of the building, corresponding to the part effectively observable in the image used



for reconstruction. This methodological choice stems from the awareness that, in a monocular process, the depth of non-visible parts cannot be determined deterministically but is instead estimated by the system on the basis of learned correlations. The distinction between global comparison and localized comparison therefore makes it possible to separate deviations attributable to the complexity of volumetric inference from discrepancies related to the restitution of morphological components directly accessible to the model.

The analysis combined quantitative parameters, such as mean error, standard deviation, and extreme values, with a qualitative morphological assessment aimed at identifying recurring modes of simplification, volumetric compression, or proportional alteration. In this way, the

deviation between the original and the reconstructed model was not interpreted solely as a metric discrepancy, but as a manifestation of the manner in which the system organizes and re-elaborates spatial complexity from a single two-dimensional input.

Results

The metric analysis performed in *Cloud-Compare* produced a Cloud-to-Mesh (C2M) distance distribution that clearly distinguishes between the global behavior of the reconstructed model and the geometric quality of the portion directly observable in the input image.

Global comparison

The comparison extended to the entire generated mesh (Fig. 6), including

Fig. 4 - Rear view of the reconstructed model of the Perugia Fontivegge railway station, corresponding to the portion not visible in the input render and therefore entirely inferred by the image-to-3D system. Source: Bianconi, F., Filippucci, M., Mommi, C. (2022).

Fig. 5 - View of the reconstructed mesh from HYPER3D-Rodin model highlighting the rear and occluded areas not visible in the input image and necessarily inferred by the image-to-3D system. Source: elaboration of Chiara Mommi.

Fig. 6 - Global Cloud-to-Mesh (C2M) signed distance map between the reconstructed model from HYPER3D-Rodin and the reference digital model, computed in *CloudCompare*. Source: elaboration of Chiara Mommi.

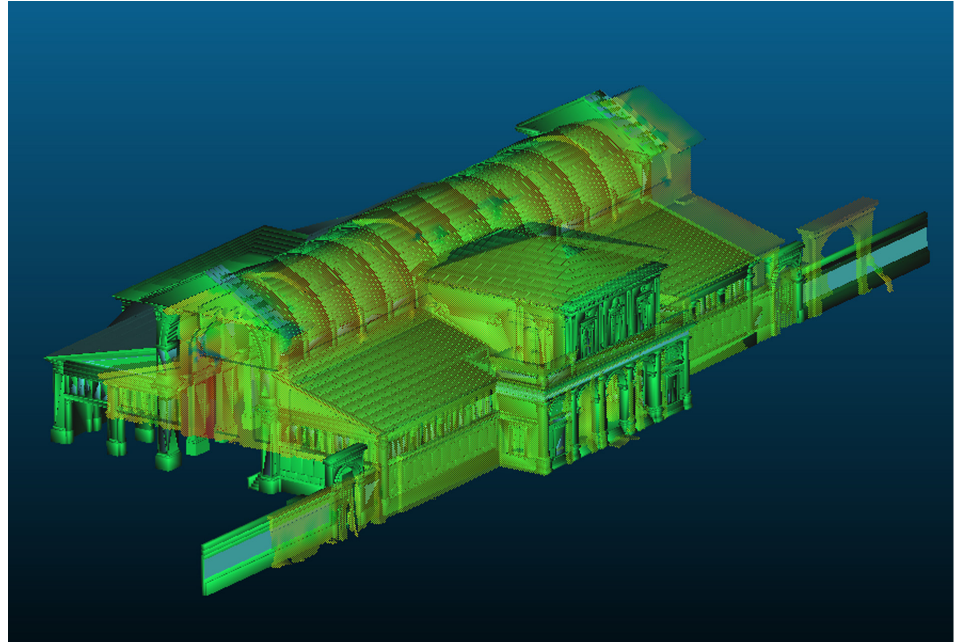
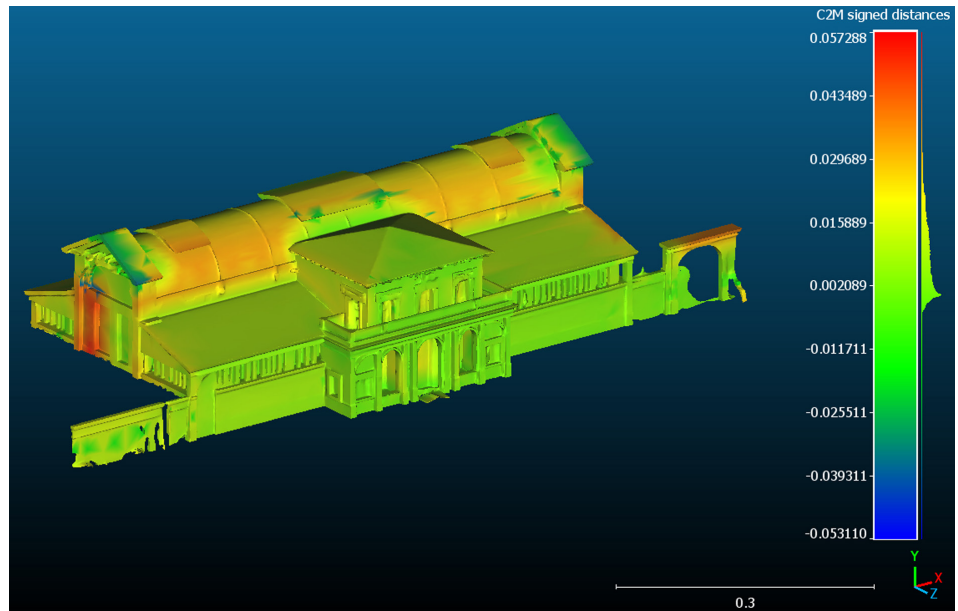


Fig. 7 - Detailed view of the signed Cloud-to-Mesh (C2M) distance distribution between the complete reconstructed model from HYPER3D-Rodin and the reference digital model, computed in *CloudCompare*. Source: elaboration of Chiara Mommi.



both the parts visible in the render and the rear portions inferred by the system, the signed distance values range between -0.0531 and $+0.0573$ (in model units). The chromatic map (Fig. 7) shows a predominance of values close to zero, indicating an overall satisfactory volumetric coherence.

At the same time, more pronounced deviations are detectable, primarily localized in areas not directly observable in the original image.

These deviations are not randomly distributed; rather, they concentrate in the posterior and lower portions of the building, namely in those areas for which the

system lacked direct visual information and was required to estimate depth and spatial configuration inferentially. The presence of extreme values in these zones confirms that monocular three-dimensional reconstruction does not consist of a neutral geometric extension, but entails an active reorganization of volumetric relationships based on learned correlations.

Overall, the global analysis outlines a condition of general coherence while simultaneously highlighting the non-deterministic character of reconstruction in non-visible regions, which significantly affects the overall magnitude of metric deviation.

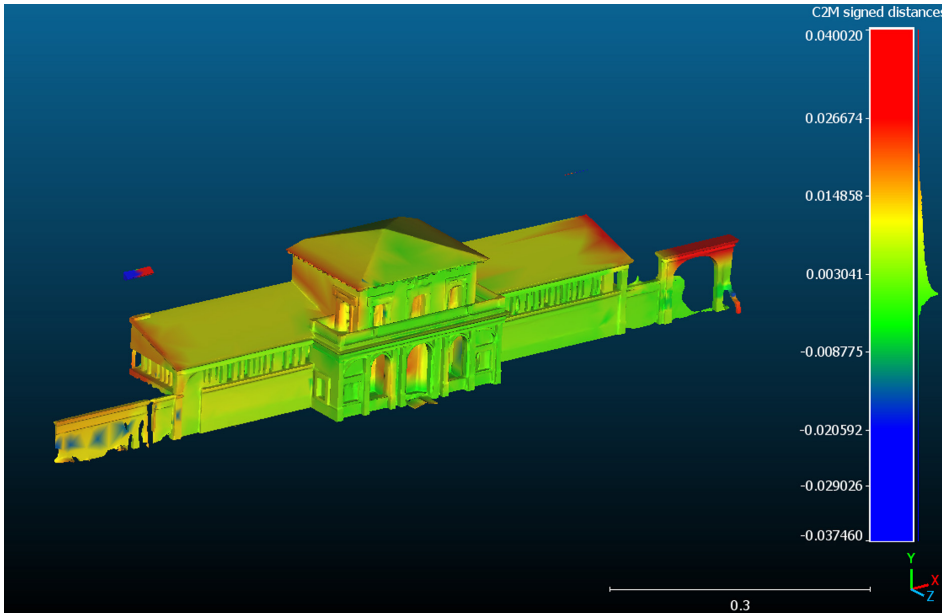


Fig. 8 - Cloud-to-Mesh (C2M) signed distance map restricted to the main façade of the reconstructed model from HYPER3D-Rodin, segmented from the complete mesh and compared with the reference digital model in *CloudCompare*. Source: elaboration of Chiara Mommi.

Localized comparison: frontal façade

To isolate the quality of reconstruction in the portions effectively accessible to the system through the input image, the analysis was repeated on the main façade alone, previously segmented from the complete mesh (Fig. 8).

In this case, the range of signed distances narrows to an interval between -0.0375 and $+0.0400$, revealing a significant contraction in the amplitude of deviations compared to the global comparison. The distribution of values is more concentrated around zero and displays substantial symmetry between positive and negative deviations, suggesting the absence of systematic deformations or unidirectional volumetric compressions.

The highest deviations are localized in correspondence with architectural elements characterized by greater morphological complexity, such as cornices, overhangs, and articulated plastic details, whereas the principal planar surfaces exhibit close adherence to the reference model. This behavior indicates that the system preserves a reasonable stability in overall proportional relationships, while tending to simplify or regularize geometrically more intricate components.

Comparative interpretation

The comparison between the global and the localized analyses reveals a substantial reduction in the amplitude of the metric range (from approximately ± 0.057 to approximately ± 0.040),

confirming that the largest deviations are predominantly associated with the portions not visible in the original image. This difference does not merely represent a numerical improvement; rather, it serves as an indicator of the system's inferential behavior. Where visual information is available, the reconstruction tends to preserve spatial relationships with greater coherence; where information is absent, the volumetric configuration is reorganized according to probabilistic logics.

The results therefore demonstrate that image-to-3D reconstruction does not produce a simple geometric replica of the design model, but instead generates a spatial configuration that maintains the overall plausibility of the architectural organism while introducing measurable variations in proportions and secondary articulations. The observed metric deviation cannot be interpreted solely as error; it should instead be understood as a quantitative manifestation of the ways in which the algorithm infers depth and structural organization from a single two-dimensional view.

Conclusion

The investigation conducted allows image-to-3D reconstruction to be interpreted not as a mere operation of geometric restitution, but as a device of algorithmic spatial translation. Consistent with the notion that every act of representation constitutes a form of mediation regulated by instruments

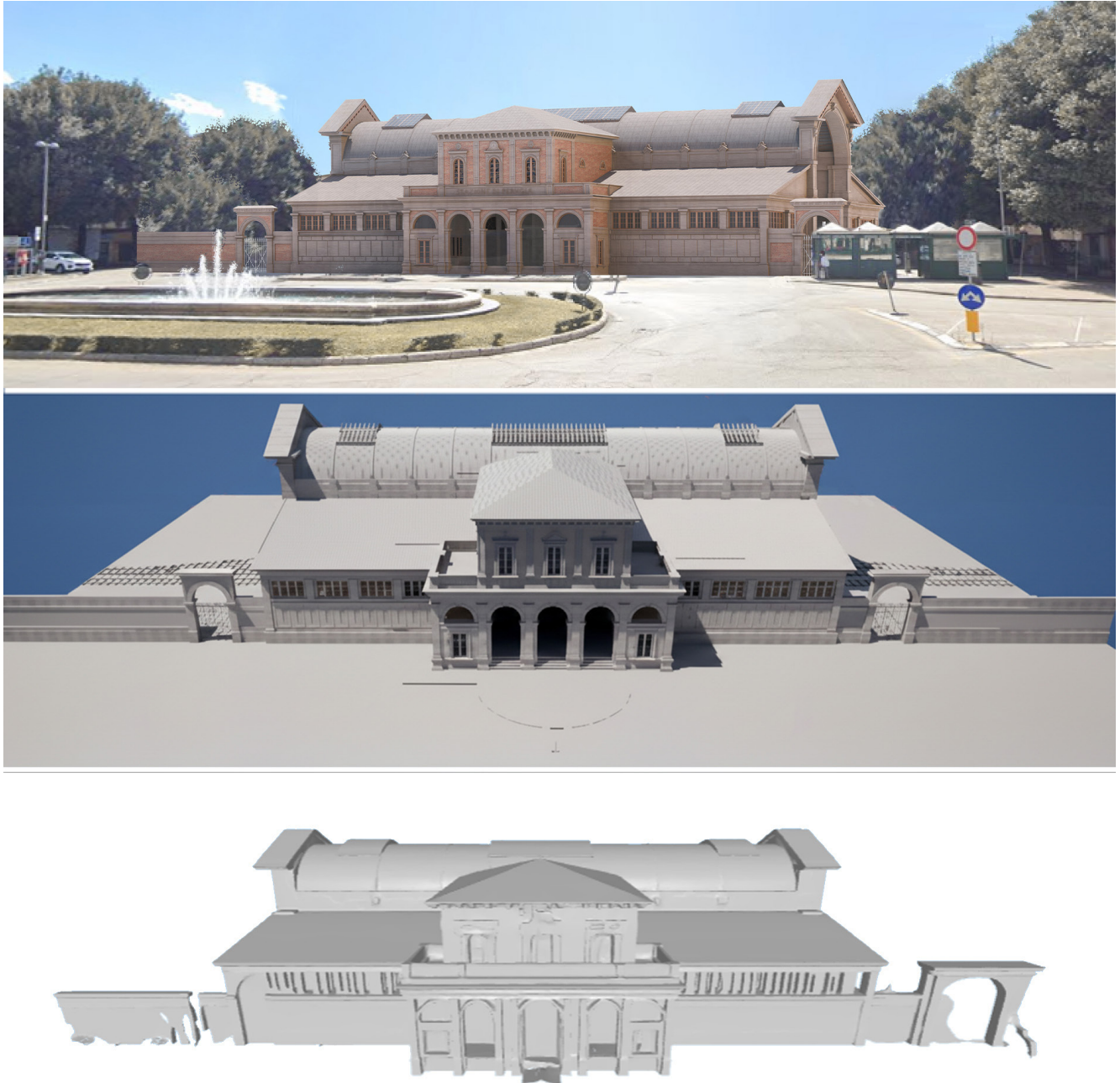


Fig. 9 - From top to bottom: photorealistic input render derived from the original digital model of the Perugia Fontivegge railway station; reference three-dimensional model; reconstructed mesh generated through the image-to-3D workflow (Hyper3D – Rodin). Source: elaboration of Chiara Mommi.

and procedures, three-dimensionality generated from a single image does not configure itself as a replica of the original model, but rather as an inferential reorganization of its volumetric relationships (Fig. 9).

The metric analysis revealed an overall satisfactory global coherence, accompanied by measurable deviations distributed differentially between directly observable portions and areas necessarily inferred by the system. The reduction in deviation range observed in the segmented façade, compared to the comparison extended to the entire mesh, confirms that where visual information is available, reconstruc-

tion maintains greater relational stability. Conversely, in non-visible zones, spatial configuration is reorganized according to probabilistic logics consistent with the statistical nature of generative neural models. From this perspective, deviation does not represent a mere metric error, but becomes an operative trace of the inferential process through which depth, proportions, and spatial relationships are deduced from partial two-dimensional data. The discrepancy thus acquires epistemic value, rendering visible the distance between explicit design configuration and algorithmically synthesized configuration.

Within this negotiated space between design and inference, the algorithmic curator (Bianconi et al., 2024), as theorized in recent reflections on AI-mediated design practices, emerges as a critical mediator capable of interpreting deviations, defining criteria of reliability, and orienting the interaction between designer and model toward a reflective understanding of architectural form. This figure coincides neither with the traditional author of geometry nor with the passive user of an automated system, but assumes a role of interpretative direction through which representation regains a renewed agency. In this sense, representation is no longer solely a descriptive instrument, but a dialogical field in which design intentionality and algorithmic synthesis are articulated. The integration between image generation through textual prompts and subsequent three-dimensional translation further opens unprecedented operational scenarios. The sequence image-model-evaluation may constitute an iterative cycle of design exploration, in which generative production is not limited to visualization

but becomes a device of spatial hypothesis. Transforming a synthetic image into a volumetric configuration enables the rapid testing of morphological alternatives, which may subsequently be subjected to metric verification, critical re-elaboration, and reintegration within controlled modeling workflows.

Within this framework, image-to-3D techniques do not replace deterministic modeling processes; rather, they complement them by introducing new modalities of conceiving and developing the project. They delineate a paradigm in which representation becomes a dynamic space of translation, inference, and negotiation, redefining methods of architectural design and expanding the operational field of formal experimentation. The value of these tools lies not exclusively in achievable metric precision, but in their capacity to make explicit the implicit logics of algorithmic spatial synthesis, thereby contributing to a renewed critical awareness of the relationship between image, model, and the construction of form.

References

- Bianconi, F., Filippucci, M., Cerbai, C., Meschini, M., Migliosi, A., & Mommi, C. (2024). Digitalizzazione e inclusione: L'intelligenza artificiale per esperienze museali multi-sensoriali. In T. Empler, A. Caldarone, & A. Fusinetti (Eds.), *DAI Il disegno per l'accessibilità e l'inclusione* (pp. 462–475). Publica.
- Bianconi, F., Filippucci, M., Migliosi, A., & Mommi, C. (2024). AI-D: Algorithmic experiments with AI to aid drawing. *SCIRES-IT*, 14(2), 31–48. <https://doi.org/10.2423/i22394303v14n2p31>
- Bianconi, F., Filippucci, M., & Mommi, C. (2022). The seduction of the simulation: 3D modelling and storytelling of unrealized Perugia rail station. *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences – ISPRS Archives*, XLIII-B2-2022, 1145–1152. <https://doi.org/10.5194/isprs-archives-XLIII-B2-2022-1145-2022>
- Blattmann, A., Dockhorn, T., Kulal, S., Mendelevitch, D., Kilian, M., Lorenz, D., Levi, Y., English, Z., Voleti, V., Letts, A., Jampani, V., & Rombach, R. (2023). *Stable video diffusion: Scaling latent video diffusion models to large datasets* (arXiv Preprint No. arXiv:2311.15127). arXiv. <https://doi.org/10.48550/arXiv.2311.15127>
- Bono, G. (2024). Text-to-building: Experiments with AI-generated 3D geometry for building design and structure generation. *Architectural Intelligence*, 3, 24. <https://doi.org/10.1007/s44223-024-00060-5>
- Carpo, M. (2011). *The alphabet and the algorithm*. The MIT Press.
- Carpo, M. (2017). The second digital turn. In *The second digital turn* (pp. 9–98). The MIT Press. <https://doi.org/10.7551/mitpress/9976.003.0004>
- Chaillou, S. (2019). *AI + architecture: Towards a new approach*. Harvard University. https://www.academia.edu/39599650/AI_Architecture_Towards_a_New_Approach
- Chang, M., Druga, S., Fiannaca, A. J., Vergani, P., Kulkarni, C., Cai, C. J., & Terry, M. (2023).

- The prompt artists. In *Proceedings of the 15th Conference on Creativity and Cognition* (pp. 75–87). <https://doi.org/10.1145/3591196.3593515>
- Choy, C. B., Xu, D., Gwak, J., Chen, K., & Savarese, S. (2016). 3D-R2N2: A unified approach for single and multi-view 3D object reconstruction. In *Computer Vision – ECCV 2016* (pp. 628–644). Springer.
- ComfyUI. (n.d.). GitHub. Retrieved September 15, 2024, from <https://github.com/comfyanonymous/ComfyUI>
- Dehouche, N., & Dehouche, K. (2023). What’s in a text-to-image prompt? The potential of stable diffusion in visual arts education. *Heliyon*, 9(6), e16757. <https://doi.org/10.1016/j.heliyon.2023.e16757>
- del Campo, M., & Manninger, S. (2022). Architecture design in the age of artificial intelligence. In C. C. Dean, M. Huppertz, & C. McWilliam (Eds.), *The Routledge companion to ecological design thinking*. Routledge.
- Dussere, C., & Mitchell, W. J. T. (1983). The language of images. *The German Quarterly*, 56(3), 477.
- Eigen, D., Puhersch, C., & Fergus, R. (2014). Depth map prediction from a single image using a multi-scale deep network. In *Advances in Neural Information Processing Systems* (Vol. 27).
- Evans, R. (1995). *The projective cast: Architecture and its three geometries*. The MIT Press.
- Girardeau-Montaut, D. (2016). *CloudCompare*. EDF R&D Telecom ParisTech.
- Godard, C., Mac Aodha, O., & Brostow, G. J. (2017). Unsupervised monocular depth estimation with left-right consistency. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 270–279).
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. The MIT Press.
- Goodman, N. (1976). *Languages of art*. Hackett.
- Ho, J., Salimans, T., Gritsenko, A., Chan, W., Norouzi, M., & Fleet, D. J. (2022). Video diffusion models. In *Advances in Neural Information Processing Systems* (Vol. 35).
- Hodder, I. (1999). *The archaeological process: An introduction*. Blackwell.
- Hunyuan3D, T., Yang, S., Yang, M., Feng, Y., Huang, X., Zhang, S., He, Z., Luo, D., Liu, H., & Zhao, Y. (2025). *Hunyuan3D 2.1: From images to high-fidelity 3D assets with production-ready PBR material* (arXiv Preprint No. arXiv:2506.15442). arXiv.
- Kerbl, B., Kopanas, G., Leimkühler, T., & Drettakis, G. (2023). 3D Gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4), 131–139.
- Liu, R., Wu, R., Van Hoorick, B., Tokmakov, P., Zakharov, S., & Vondrick, C. (2023). Zero-1-to-3: Zero-shot one image to 3D object. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)* (pp. 9298–9309). <https://doi.org/10.1109/ICCV51070.2023.00856>
- Meshy AI. (2023). *Meshy AI: Image-to-3D generation platform*. <https://www.meshy.ai>
- Mildenhall, B., Srinivasan, P. P., Tancik, M., Barron, J. T., Ramamoorthi, R., & Ng, R. (2020). NeRF: Representing scenes as neural radiance fields for view synthesis. In *Lecture Notes in Computer Science* (Vol. 12346 LNCS, pp. 405–421). Springer. <http://arxiv.org/abs/2003.08934>
- Poole, B., Jain, A., Barron, J. T., & Mildenhall, B. (2022). *DreamFusion: Text-to-3D using 2D diffusion* (arXiv Preprint No. arXiv:2209.14988). arXiv. <https://doi.org/10.48550/arXiv.2209.14988>
- Qin, S., Xiao, J., & Ge, J. (2024). Dip-NeRF: Depth-based anti-aliased neural radiance fields. *Electronics*, 13(8), 1527. <https://doi.org/10.3390/electronics13081527>
- Rombach, R., Blattmann, A., Lorenz, D., Esser, P., & Ommer, B. (2022). High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 10684–10695). <https://doi.org/10.1109/CVPR52688.2022.01042>
- Tatarchenko, M., Dosovitskiy, A., & Brox, T. (2015). *Single-view to multi-view: Reconstructing unseen views with a convolutional network*. arXiv. <https://arxiv.org/abs/1511.06702>
- Tochilkin, D., Pankratz, D., Liu, Z., Huang, Z., Letts, A., Li, Y., Liang, D., Laforte, C., Jampani, V., & Cao, Y.-P. (2024). *TripoSR: Fast 3D object reconstruction from a single image* (arXiv Preprint No. arXiv:2403.02151). arXiv.

- Tulsiani, S., Zhou, T., Efros, A. A., & Malik, J. (2017). Multi-view supervision for single-view reconstruction via differentiable ray consistency. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 2626–2634).
- Wang, S., & Zhang, Y. (2024). 3DSAM: Segment anything in NeRF. In *ICASSP 2024 – 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 3060–3064). IEEE. <https://doi.org/10.1109/ICASSP48485.2024.10445897>
- Wang, T., Zhang, B., Zhang, T., Gu, S., Bao, J., Baltrusaitis, T., Shen, J., Chen, D., Wen, F., & Chen, Q. (2023). Rodin: A generative model for sculpting 3D digital avatars using diffusion. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 4563–4573).
- Weiss, C., & Beskardes, G. (2025). *HiTEM*. Sandia National Laboratories (SNL-NM), Albuquerque, NM, United States.
- Wu, J., Zhang, C., Xue, T., Freeman, B., & Tenenbaum, J. (2016). Learning a probabilistic latent space of object shapes via 3D generative-adversarial modeling. In *Advances in Neural Information Processing Systems* (Vol. 29).
- Xia, B., Zhang, Y., Li, J., Wang, C., Wang, Y., Wu, X., Yu, B., & Jia, J. (2025). DreamOmni: Unified image generation and editing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 28533–28543). <https://doi.org/10.1109/CVPR52734.2025.02657>
- Zhang, L., Rao, A., & Agrawala, M. (2023). Adding conditional control to text-to-image diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)* (pp. 3813–3824). <https://doi.org/10.1109/ICCV51070.2023.00355>
- Zheng, G., Li, T., Jiang, R., Lu, Y., Wu, T., & Li, X. (2024). *CamI2V: Camera-controlled image-to-video diffusion model* (arXiv Preprint No. arXiv:2410.15957). arXiv. <https://doi.org/10.48550/arXiv.2410.15957>